Introduction to University Mathematics

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Autumn 2026

Abstract

Notes for the University of Helsinki course MAT11001, taught in Finnish from 31.08.2026–17.10.2026.

Contents

1 Course information

The official course information is available on the University of Helsinki course page.

2 Lecture notes

2.1 31.08.2026

The first lecture was mostly spent going over the course structure and other general admin topics. We did go over some preliminary definitions and examples, which will follow.

2.1.1 Notational conventions

Definition 2.1. An integer $p > 1$ is *prime* if its only positive divisors are 1 and p.

Definition 2.2. The following basic sets of numbers will be used throughout:

$$\begin{aligned}
\mathbb{N} &= \{0, 1, 2, \ldots\} \\
\mathbb{Z} &= \{\ldots, -2, -1, 0, 1, 2, \ldots\} \\
\mathbb{Z}^+ &= \mathbb{N} \setminus \{0\} \\
\mathbb{Q} &= \left\{ \frac{p}{q} : p, q \in \mathbb{Z},\ q \neq 0 \right\} \\
\mathbb{R} &= \text{the set of real numbers} \\
\mathbb{C} &= \{a + bi : a, b \in \mathbb{R}\}, \quad i^2 = -1 \\
\mathbb{P} &= \{p \in \mathbb{Z} : p \text{ is prime}\}
\end{aligned}$$

2.1.2 Principle of mathematical induction

Here we state the fundamental principle of mathematical induction without proof, and use it freely thereafter.

Theorem 2.3 (Principle of mathematical induction)**.** *Let $n_0 \in \mathbb{N}$, and let $P(n)$ be a statement for each $n \in \mathbb{N}$ with $n \geq n_0$. Suppose that*

1. *$P(n_0)$ is true, and*
2. *for every $n \geq n_0$, $P(n)$ implies $P(n+1)$.*

Then $P(n)$ is true for every $n \geq n_0$.

Proof. Omitted. □

Example 2.4. We prove by induction that

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$$

for every integer $n \geq 1$.

For $n = 1$, the identity holds because

$$\sum_{k=1}^{1} k = 1 = \frac{1(1+1)}{2}.$$

Now suppose that the identity holds for some integer $n \geq 1$. Then

$$\begin{aligned}
\sum_{k=1}^{n+1} k &= \sum_{k=1}^{n} k + (n+1) \\
&= \frac{n(n+1)}{2} + (n+1) \\
&= \frac{(n+1)(n+2)}{2}.
\end{aligned}$$

Thus the identity also holds for $n + 1$, and hence for every integer $n \geq 1$. □

2.2 03.09.2026

2.2.1 Divisibility of integers

We start by defining the precise meaning of *even* and *odd* numbers.

Definition 2.5. A number $m \in \mathbb{Z}$ is *even* if there exists $k \in \mathbb{Z}$ such that

$$m = 2k.$$

On the other hand, m is *odd* if there exists a $k \in \mathbb{Z}$ such that

$$m = 2k + 1.$$

Definition 2.6 (Divisibility)**.** A number $a \in \mathbb{Z}$ is said to be *divisible* by a number $b \in \mathbb{Z}, b \neq 0$, if there exists a $k \in \mathbb{Z}$ such that

$$a = kb.$$

In this case we also say that b *divides* a, and write

$$b \mid a.$$

If b does *not* divide a, we write

$$b \nmid a.$$

Definition 2.7. For $n \in \mathbb{Z} \setminus \{0\}$, we denote the set of integers that are divisible by n by

$$n\mathbb{Z} = \{z \in \mathbb{Z} : z = nk, \quad k \in \mathbb{Z}\}.$$

2.2.2 Strong induction

In some proofs it is useful to assume all preceding cases rather than only the immediately preceding case. This gives an equivalent form of the principle of induction, called *strong induction*.

Theorem 2.8 (Strong induction)**.** *Let $n_0 \in \mathbb{N}$, and let $P(n)$ be a statement for each $n \in \mathbb{N}$ with $n \geq n_0$. Suppose that*

1. *$P(n_0)$ is true, and*
2. *for every $n \geq n_0$, if $P(k)$ is true for every k with $n_0 \leq k \leq n$, then $P(n+1)$ is true.*

Then $P(n)$ is true for every $n \geq n_0$.

Proof. Omitted. □

As an example where strong induction is natural, let us prove that all natural numbers $n \geq 2$ can be written as a product of *prime* numbers.

Theorem 2.9. *Every natural number $n \geq 2$ can be written as a product of prime numbers.*

Proof. When $n = 2$, we have $n \in \mathbb{P}$, so the result holds trivially. Now suppose $n \geq 2$ and the statement holds for all $2 \leq k \leq n$. If $n+1 \in \mathbb{P}$, the result holds trivially, so suppose $n + 1 \notin \mathbb{P}$. Then we have

$$n + 1 = ab,$$

for some $2 \leq a, b \leq n$. By the induction hypothesis, a and b can be written as products of primes, so

$$\begin{aligned} a &= p_1 p_2 \cdots p_i, \quad p_1, \ldots, p_i \in \mathbb{P}, \\ b &= q_1 q_2 \cdots q_j, \quad q_1, \ldots, q_j \in \mathbb{P}. \end{aligned}$$

Thus we have

$$n + 1 = ab = p_1 p_2 \cdots p_i q_1 q_2 \cdots q_j.$$

The result follows from Theorem 2.8. $\square$

2.2.3 Sequences

Definition 2.10. A sequence of real numbers is a function $z \colon \mathbb{N} \to \mathbb{R}$, whose terms are denoted by $z_n = z(n)$. We write the sequence as

$$(z_n)_{n \in \mathbb{N}} = (z_0, z_1, z_2, \ldots).$$

Definition 2.11. A sequence of real numbers may also be specified recursively. First, for some $k \in \mathbb{Z}^+$, we define the initial terms $z_0, \ldots, z_{k-1}$. For each $n \geq k$, we then define z_n in terms of earlier terms $z_0, \ldots, z_{n-1}$.

2.3 07.09.2026

This lecture was mostly spent going through basic mathematical notation.

2.3.1 Logical connectives

Definition 2.12. Let p and q be propositions. We define the following logical connectives:

1. The *conjunction* $p \wedge q$, read "p and q", is true exactly when both p and q are true.

2. The *disjunction* $p \vee q$, read "p or q", is true exactly when at least one of p and q is true.

3. The *implication* $p \implies q$, read "p implies q", is false exactly when p is true and q is false.

4. The *equivalence* $p \iff q$, read "p if and only if q", is true exactly when p and q have the same truth value.

5. The *negation* $\neg p$, read "not p", is true exactly when p is false.

Definition 2.13. Let p and q be propositions.

1. The *negated implication* $p \not\Rightarrow q$, read "p does not imply q", is true exactly when p is true and q is false.

2. The *negated equivalence* $p \not\Leftrightarrow q$, read "p is not equivalent to q", is true exactly when p and q have different truth values.

Definition 2.14. Let X be a set. A *predicate* $P(x)$ on X is a statement containing a variable x whose truth value depends on the chosen value of $x \in X$. For each $a \in X$, substituting a for x gives the proposition $P(a)$, which is either true or false. For example, $P(x) \coloneqq x > 0$ is a predicate on $\mathbb{R}$; $P(2)$ is true, whereas $P(-1)$ is false.

Definition 2.15. Let $P(x)$ be a predicate on a set X.

1. The *universal quantification* $\forall x \in X \colon P(x)$, read "for every $x \in X$, $P(x)$", is true exactly when $P(a)$ is true for every $a \in X$.

2. The *existential quantification* $\exists x \in X \colon P(x)$, read "there exists an $x \in X$ such that $P(x)$", is true exactly when $P(a)$ is true for at least one $a \in X$.

Definition 2.16. The *negated existential quantification* $\nexists x \in X \colon P(x)$, read "there does not exist an $x \in X$ such that $P(x)$", is true exactly when $P(a)$ is false for every $a \in X$. Equivalently,

$$\nexists x \in X \colon P(x) \qquad \Longleftrightarrow \qquad \forall x \in X \colon \neg P(x).$$

2.3.2 Basic set theory

Definition 2.17. A *set* is a collection of distinct objects, called its *elements*. If an object x is an element of a set A, we write $x \in A$; otherwise, we write $x \notin A$. A set is determined entirely by its elements: two sets are equal exactly when they have the same elements. Thus the order in which the elements are listed, and any repetition among them, do not affect the set.

Definition 2.18. The *empty set* is the set containing no elements. It is denoted by $\emptyset$.

Definition 2.19. Let A and B be are sets. A and B are said to be *disjoint*, if $A \cap B = \emptyset$.

Definition 2.20. A *singleton* set is a set consisting of exactly **one** element.

Definition 2.21. Let $P(x)$ be a predicate on a set X. The *set-builder notation*

$$A \coloneqq \{x \in X \colon P(x)\}$$

defines A to be the set of precisely those elements $x \in X$ for which $P(x)$ is true. Equivalently, for every $a \in X$,

$$a \in A \iff P(a).$$

Definition 2.22. A set A is a *subset* of a set B if every element of A is also an element of B. We then write $A \subseteq B$. If A is not a subset of B, we write $A \nsubseteq B$; this means that some element of A is not an element of B. If $A \subseteq B$ and $A \neq B$, then A is a *proper subset* of B, written $A \subset B$.

Definition 2.23. Let A and B be subsets of a set X.

1. The *union* of A and B is

$$A \cup B \coloneqq \{x \in X \colon x \in A \vee x \in B\}\,.$$

2. The *intersection* of A and B is

$$A \cap B \coloneqq \{x \in X \colon x \in A \wedge x \in B\}\,.$$

3. The *difference* of A and B is

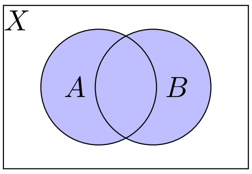
$$A \setminus B \coloneqq \{x \in X \colon x \in A \wedge x \notin B\}\,.$$

4. The *complement* of A in X is

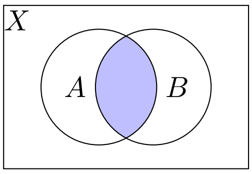
$$\complement A \coloneqq X \setminus A = \{x \in X \colon x \notin A\}\,.$$

5. The *symmetric difference* of A and B is

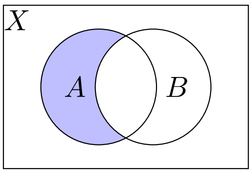
$$A \,\Delta\, B \coloneqq (A \setminus B) \cup (B \setminus A).$$



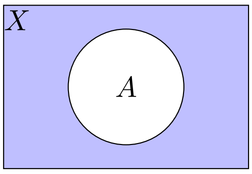
$A \cup B$



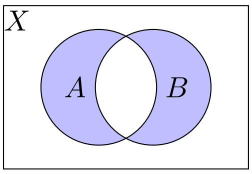
$A \cap B$



$A \setminus B$



$\complement A$



$A \,\Delta\, B$